The maintenance of electrochemical gradients in biologic systems is so important that it consumes perhaps 30–40% of the total energy expenditure in a cell. In general, cells maintain a low intracellular $Na^+$ concentration and a high intracellular $K^+$ concentration, along with a net negative electrical potential inside. The pump that maintains these gradients is an ATPase that is activated by $Na^+$ and $K^+$ ($Na^+$-$K^+$ ATPase;). The ATPase is an integral membrane protein and requires phospholipids for activity. The ATPase has catalytic centers for both ATP and $Na^+$ on the cytoplasmic side of the membrane, but the $K^+$ binding site is located on the extracellular side of the membrane. These pumps are particularly abundant in nerve cells, which are constantly pumping out sodium ions and pulling in potassium ions to maintain an electrical gradient across their cell membranes. An electrical gradient is a difference in electrical charge across a space. In the case of nerve cells, for example, the electrical gradient exists between the inside and outside of the cell, with the inside being negatively-charged (at around -70 mV) relative to the outside. The negative electrical gradient is maintained because each $Na^+/K^+$ pump moves three $Na^+$ ions out of the cell and two $K^+$ ions into the cell for each ATP molecule that is used . This process is so important for nerve cells that it accounts for the majority of their ATP usage.

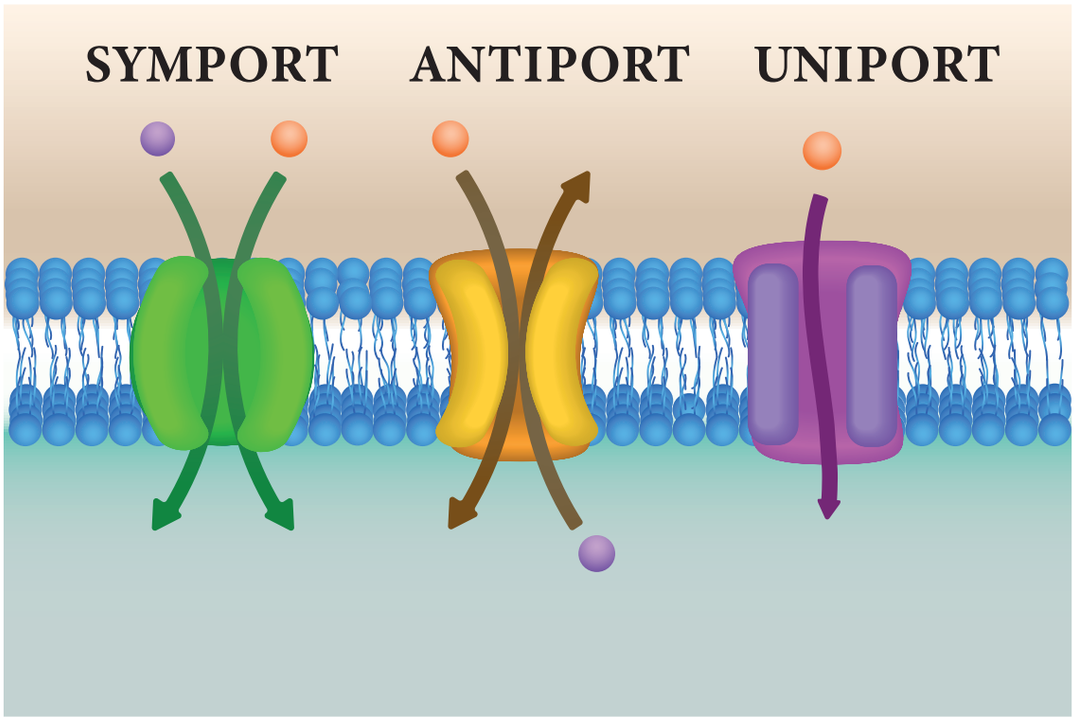



**Figure 6: Active transport across membranes. Primary reaction is mediated by pumps (see text) while the porters (Uniporter, symporter and Antiporter) are referred as secondary reactions**

Like pumps, carriers are found in all membranes, for exchange of molecules for metabolism or expelling out the wastes. Carriers are also known as facilitators or porters and the substrates for carriers include ions and small soluble organic molecules and sometimes, lipid soluble substances.



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